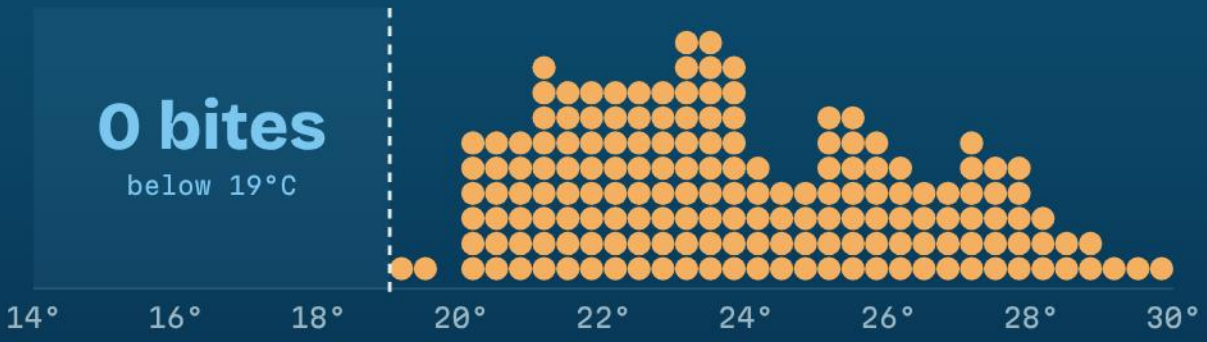
ONE RULE, 126 YEARS

Bull sharks have never bitten in water below 19°C.



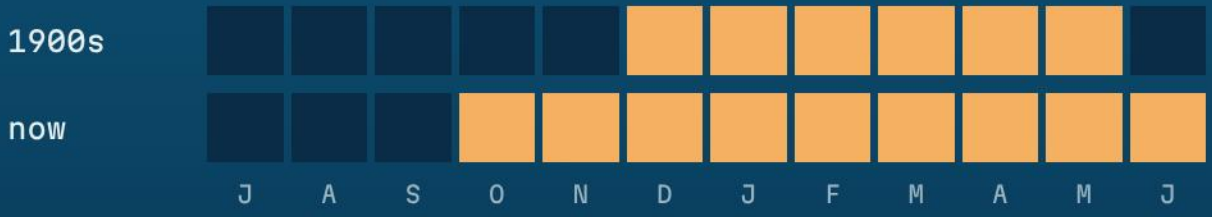
Each dot is one of 170 bites in Australia since 1900, placed at that month's sea temperature. Here's what's happening to that water.

19 Degrees · © 2026 Blake Erickson

Swipe →

MORE TIME

Off Sydney, that water now lasts 45% longer each year.



The sea is 19°C or warmer about 9 months a year, up from 6 a century ago.

MORE COASTLINE

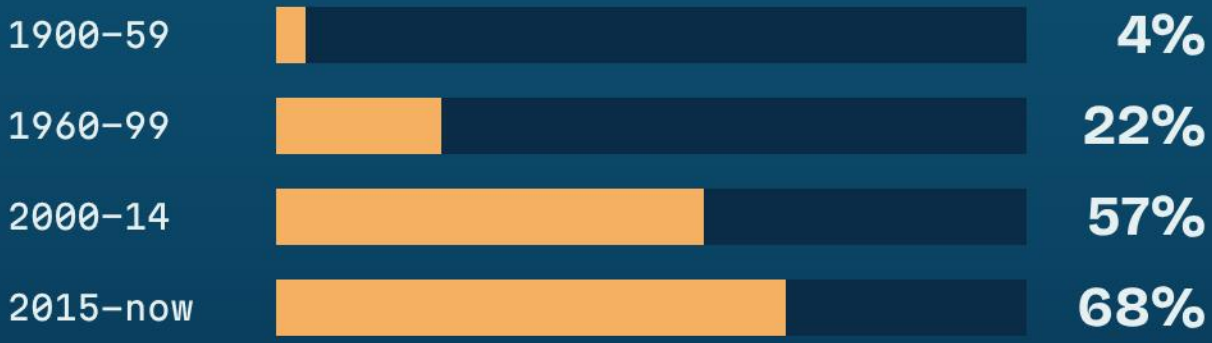
And it now reaches 215 km further down the coast.



Where the sea stays 19°C or warmer for half the year: once near Jervis Bay, now near Eden.

SO THE BITES MOVED

Out of the rivers, into coastal waters.



68% of Australia's bull shark bites since 2015 were in coastal waters, up from 4%. South of Port Macquarie, nearly half now come in autumn and winter.

THE TWIST

There are 71% fewer sharks than in 1970.



Oceanic sharks and rays, worldwide. Yet since 2020, Western Australia has had more bull shark bites (10) than in the 120 years before that combined (9). It isn't more sharks. It's where and when we meet them.

19 Degrees · © 2026 Blake Erickson

Swipe →

IT'S NOT JUST AUSTRALIA

Ocean life is moving toward the poles 72 km every decade.

12 times faster than life on land. Tiger sharks off the US travel about 400 km further north for every 1°C of warming. The ocean has set a heat record nine years running.

KEEP IT IN PERSPECTIVE

For every shark death, 49 people drown.

The biggest danger at the beach is still the water itself.

 Swim between the red and yellow flags. Stay out of murky water for a few days after heavy rain. Check the SharkSmart app.

EXPLORE IT YOURSELF

Free and open source.

Every chart, method and number, with downloadable data and classroom sheets. Researchers: test the claims, add data and improve the methods. Link in the comments. What would you test next?